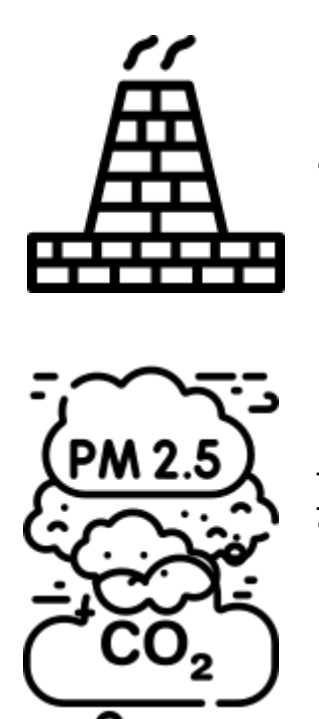


Brick Kilns Across India: A Temporal Analysis of Brick Kiln Distribution and Technology Types

Sugandha Arora¹, Clement Goldmann², Stefan Oehmcke³, Jan Pauls¹, Chuanlong Zhou², Kushal Tibrewal², Harish Phuleria⁴, Philippe Ciais², Fabian Gieseke¹
¹University of Münster, Münster, Germany. ²Laboratoire des Sciences du Climat et de l'Environnement, IPSL CEA CNRS UVSQ, Gif-sur-Yvette, France. ³University of Rostock, Germany, ⁴Indian Institute of Technology Bombay, India.

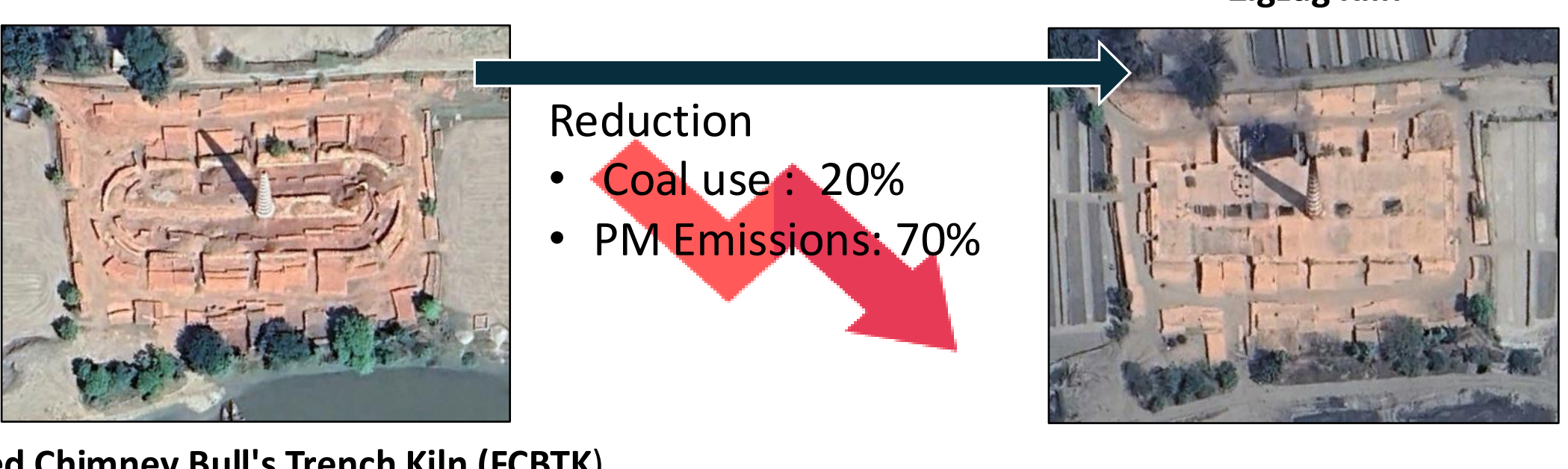
1 Introduction

Sector Overview



- This sector produces **233 ± 15 billion** brick/year, in India^[1] and pollution associate **127Mt/year of CO2**
- A total of **100,000** brick kilns are estimated in India, including **55,000** in the « brick belt »^[2] and **~170 kt PM_{2.5}/year 15 % national total**^[3]

Technology Switch



Reduction

- Coal use : 20%
- PM Emissions: 70%

Fixed Chimney Bull's Trench Kiln (FCBTK) → Zigzag Kiln

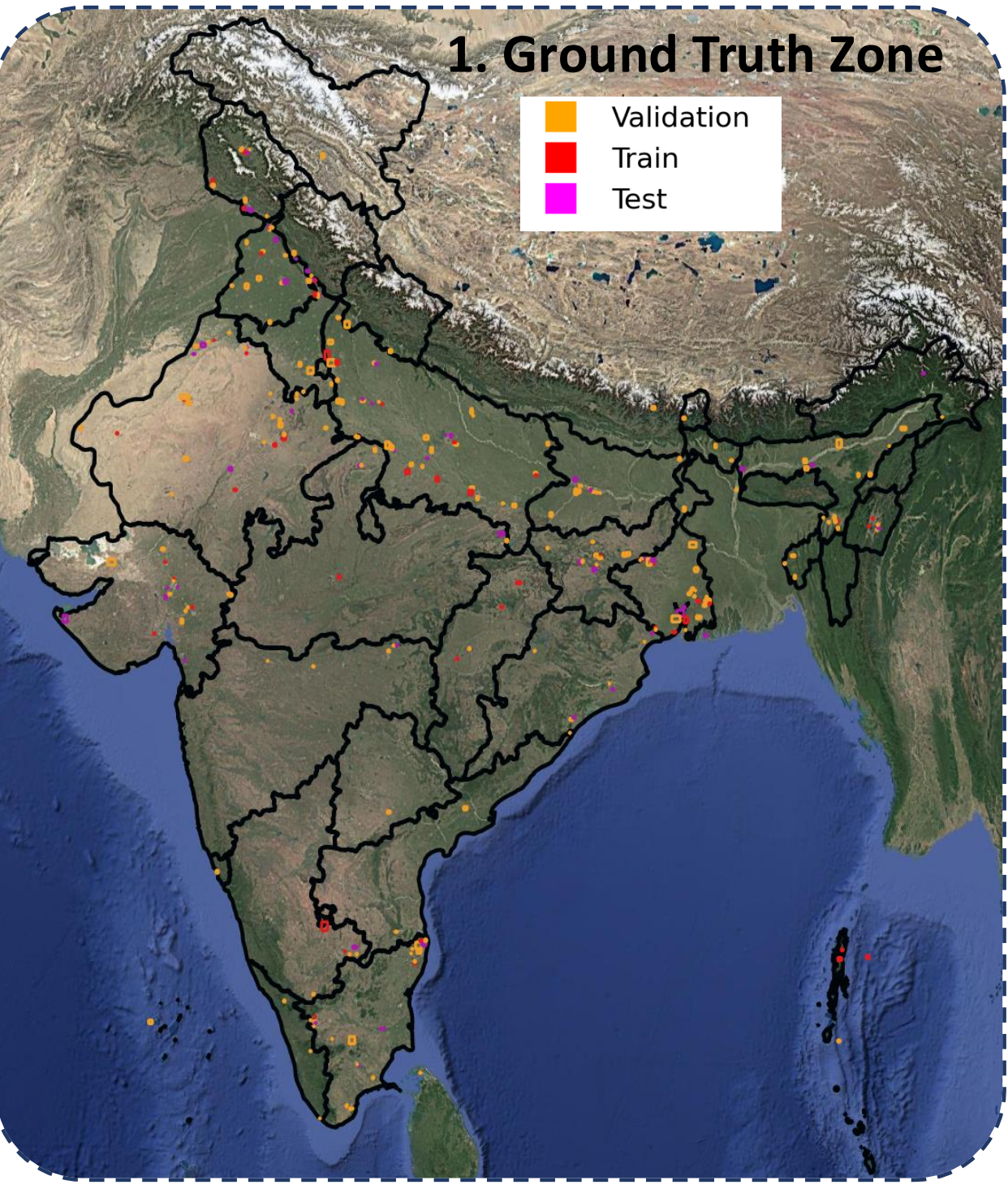
Objectives :

- Mapping brick kiln locations and technology type classification across India
- Temporal Evolution of Brick Kilns spread across India

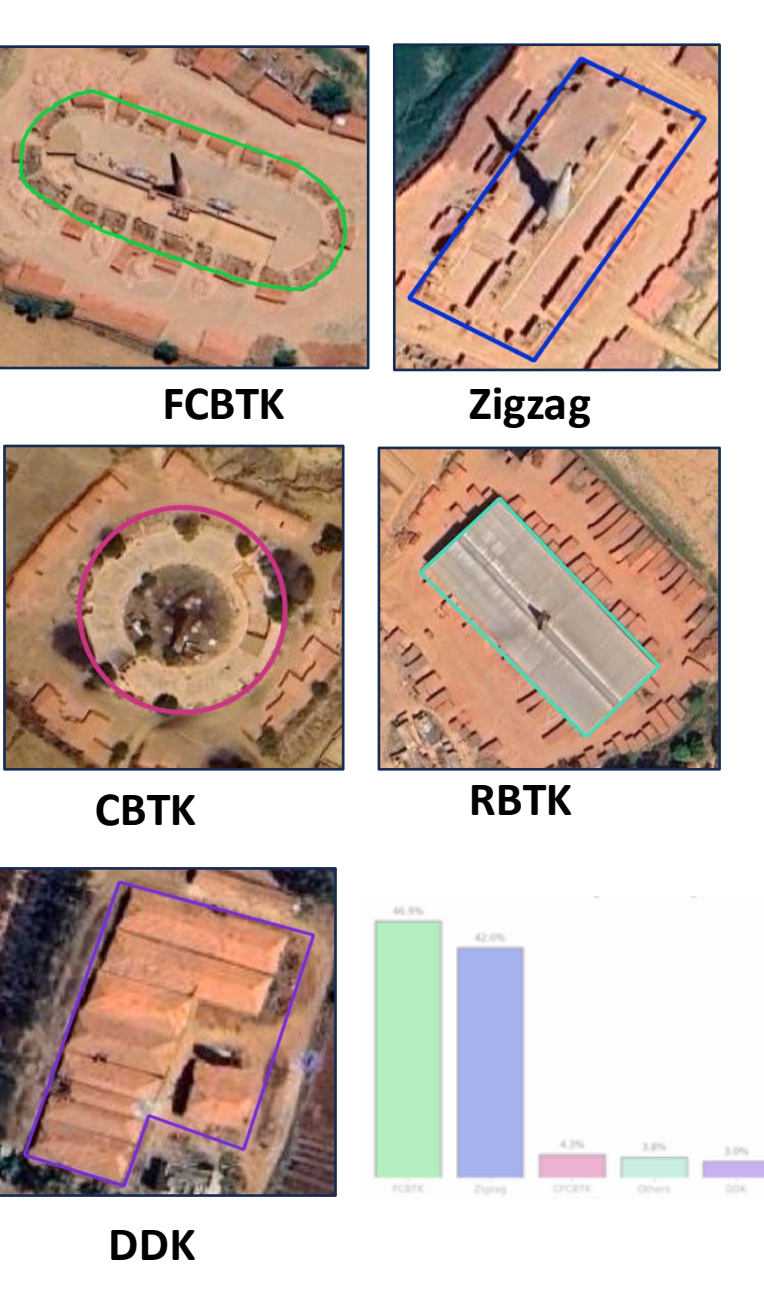
2 Methodology

Data Curation

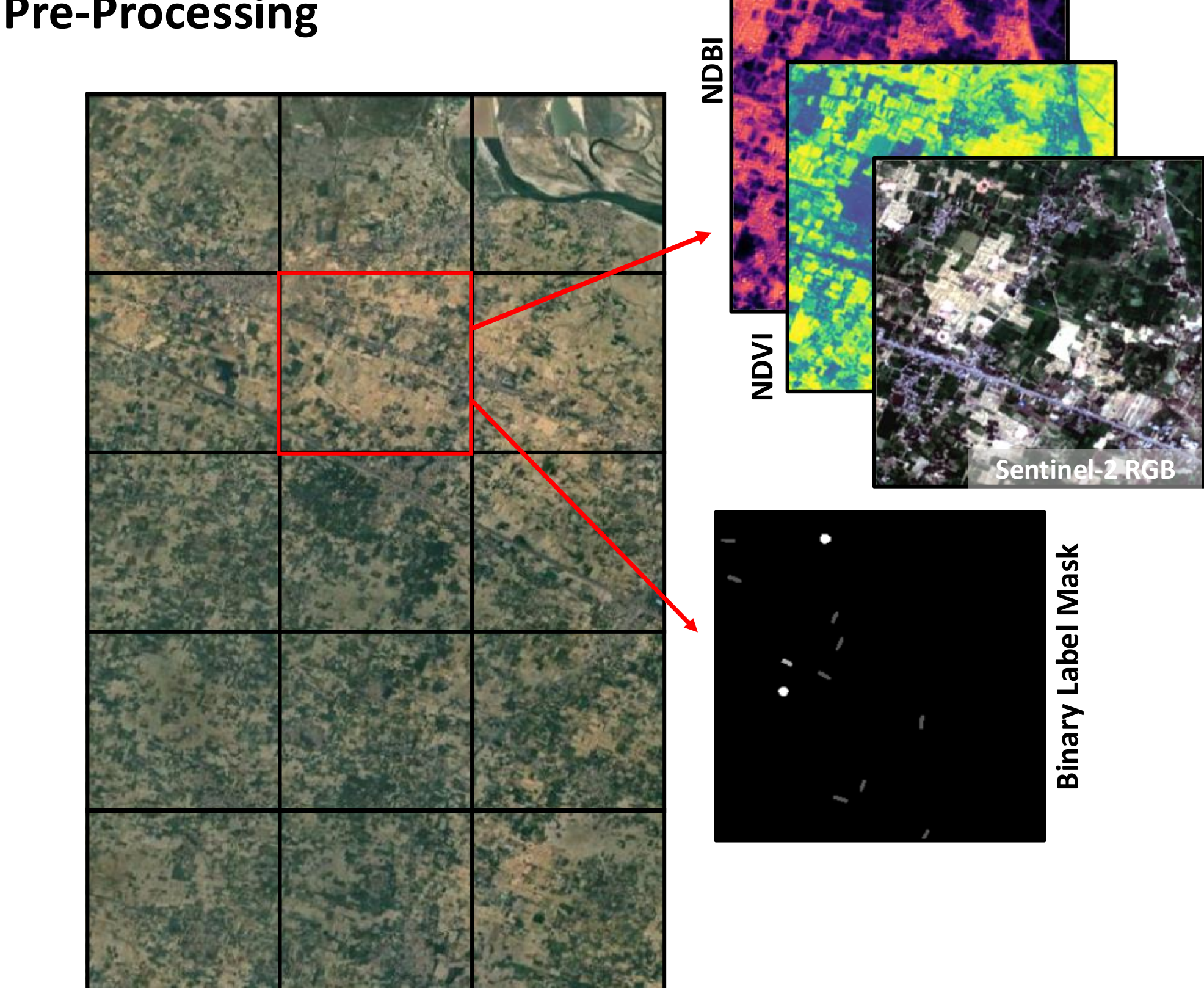
1. Ground Truth Zone



2. Ground Truth Label

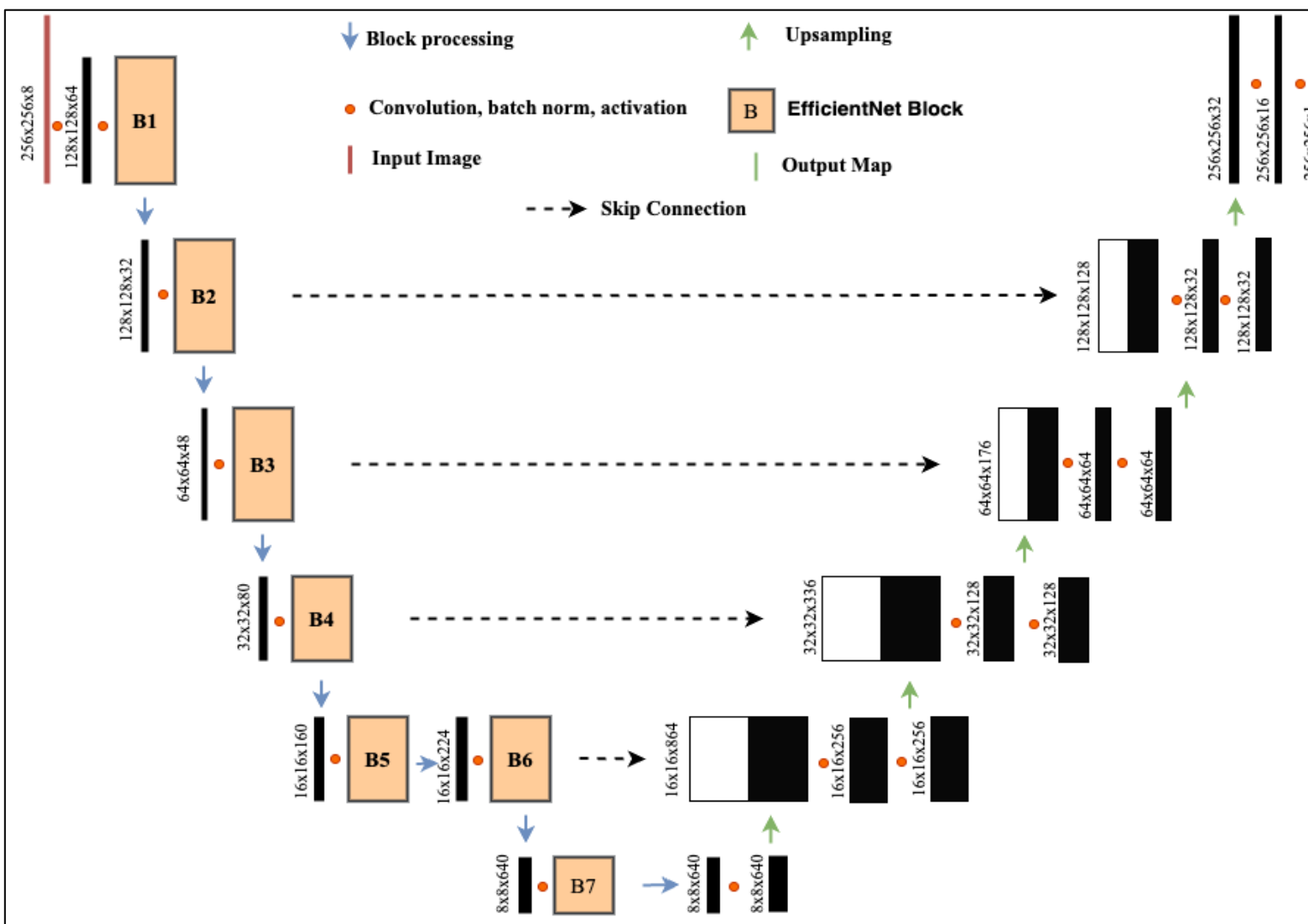


Pre-Processing

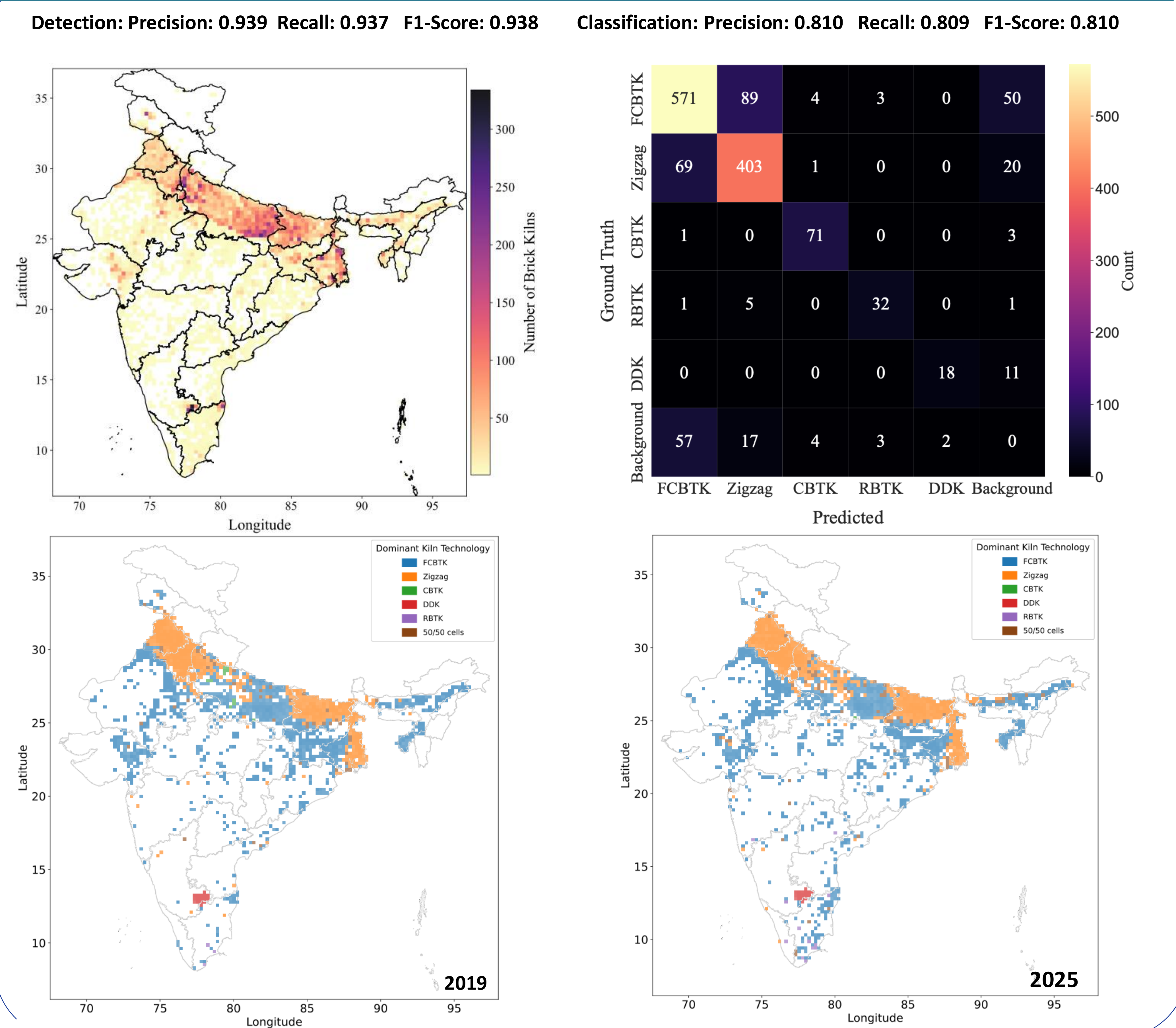


Experimental Details

- UNet with Efficient Net backbone
- Cross Entropy loss function with dynamic penalties at every epoch



3 Results



4 Conclusion and Limitation

- Provides the first nationwide multi-year brick kiln maps of India (2019-2025) using 10m resolution satellite imagery
- introduces **MuTBriK**, a large annotated dataset covering five kiln types
- trained a segmentation model with a tailored confusion-aware loss that separates visually similar classes, yielding 93% detection and 81 % classification accuracy (+ 5% over baseline).
- Detecting:
 - 42,928 kilns in the year 2025.
 - Zigzag kilns: 22157 to 23018, and
 - FCBTK : 23228 to 22851 since 2019
- Temporal analysis highlights a growing adoption of cleaner Zigzag kiln technology over time

5 Data Availability

The dataset and the nationwide brick kilnmaps supporting the findings of this study are publicly available on Zenodo at <https://zenodo.org/records/18301279>

 **Sugandha**
Email : ssugandh@uni-muenster.de

References

[1] Tibrewal, K et al. 2023. Reconciliation of energy use disparities in brick production in India. DOI : [10.1038/s41893-023-01165-x](https://doi.org/10.1038/s41893-023-01165-x).

[2] Rajarathnam, et al. 2014. Assessment of air pollutant emissions from brick kilns. DOI : [10.1016/j.atmosenv.2014.08.075](https://doi.org/10.1016/j.atmosenv.2014.08.075).

[3] Tibrewal K, et al 2019. A state-level methodology for estimating present-day emissions of short-lived climate pollutants from fired brick production in India. DOI : [10.1007/978-981-13-0280-0_13](https://doi.org/10.1007/978-981-13-0280-0_13).